WasteSort

Basic AI and machine learning

Monday learning task



Goal Diggers
BCS34 October 21, 2026

Artificial intelligence and machine learning

ARTIFICIAL INTELLIGENCE

The broad field of systems that perform tasks such as recognition and decision-making.

MACHINE LEARNING

An approach within AI that learns patterns from data to make predictions.

Deep learning uses multilayer neural networks within machine learning.

Common learning types

Learning type	What the model learns from	Example
Supervised	Examples with known labels	Photo and its material label
Unsupervised	Data without target labels	Groups of similar customers
Reinforcement	Actions and reward feedback	Game-playing strategy

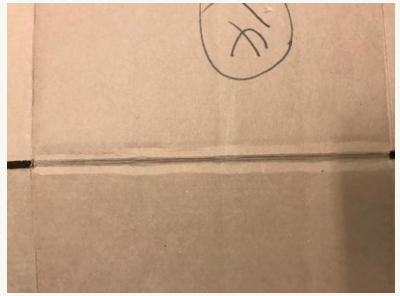
WasteSort uses supervised learning.

Practical machine learning applications

Spam filtering	Predict an email category Classification
Travel-time estimation	Predict a duration in minutes Regression
Customer grouping	Find similar examples Clustering

The WasteSort problem

One item photo. One predicted material category.



Cardboard



Glass



Metal



Paper



Plastic



Trash

TrashNet: 2,527 original labeled photos by Thung and Yang

Supervised image classification in WasteSort

1 Training

Labeled photos teach the model patterns associated with each material.

2 Evaluation

Separate images help check whether the model can classify unseen examples.

3 Prediction

A saved model predicts a material for a new uploaded photo.

What WasteSort can support

Recognize material appearance

A supervised model learns from labeled examples and predicts a category.

Use the result with care

Predictions can be wrong. Campus bin guidance requires verified local rules.